The square maximises the ratio of the first two Dirichlet eigenvalues among quadrilaterals

Anonymous

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Abstract

(draft) Let $\lambda_1(Q) < \lambda_2(Q)$ be the first two Dirichlet eigenvalues of the Laplacian on a bounded planar domain Q and $\xi(Q) = \lambda_2(Q)/\lambda_1(Q)$. We prove that $\xi(Q) \le 5/2$ for every simple quadrilateral Q, convex or not, with equality only for squares. This is the $k = 4$ case of the polygonal Payne–Pólya–Weinberger conjecture of Antunes and Freitas; the case $k = 3$ (equilateral triangle, ratio $7/3$) was proved by Siudeja and by Arbon, Mannan, Psenka and Ragavan. The proof is computer-assisted. (i) Two analytic lemmas remove near-triangles and thin shapes (width/diameter $\le 1/10$) and confine the problem to a compact parameter set. (ii) A certified local step establishes the inequality on an explicit neighbourhood of the square, via an inertia criterion for a 3×3 Rayleigh–Ritz pencil of transported eigenfunctions and a Kato–Temple bound expanded as Taylor models. (iii) Certified box covers of the remaining parameter set bound λ_2 by Rayleigh–Ritz and λ_1 by a Lehmann–Goerisch inequality with an exactly divergence-matched flux; away from the square, the test functions and the flux are transported from the box centre by a piecewise-affine map and a contravariant Piola transform, so that all parameter dependence is explicit. (iv) The band $1/10 <$ width/diameter $\le 2/5$ is covered in diameter-normalised charts by one-dimensional bounds: a Born–Oppenheimer lower bound for λ_1 certified by Sturm oscillation, and a weighted one-dimensional Ritz upper bound for λ_2. All certificates are computed in ball arithmetic (Arb) or exact rational arithmetic; every cover is replayed by a fail-closed checker. For convex quadrilaterals, coverage, exclusions and samples of certificates are re-checked by independently written programs; for non-convex ones, coverage is. (v) Non-convex quadrilaterals are handled in a separate chart centred on the reflex vertex. Analytic lemmas cover the degenerate and unbounded parts of that shape space: shallow notches, thin wedges, flat hulls and thin stems, and long arms through Dirichlet–Neumann bracketing and one-dimensional bounds with exponential tails. Certified covers of compact families, in charts uniform in the arm lengths, cover the rest.

1. Introduction and statement

Theorem 1. For every simple quadrilateral $Q \subset \mathbb{R}^2$, convex or not, $\lambda_2(Q)/\lambda_1(Q) \le 5/2$, with equality if and only if Q is a square.

Proof. A simple quadrilateral is either convex or has exactly one reflex vertex (a dart). For convex Q this is Theorem $1'$ (§4.6); for darts, Theorem 5 (§5) gives the strict inequality. □

Context. Payne, Pólya and Weinberger (1956) proved $\xi \le 3$ for all planar domains and conjectured that the disk is the maximiser ($\xi = j_{1,1}^2/j_{0,1}^2 \approx 2.539$); Ashbaugh and Benguria proved this in 1991–92. Antunes and Freitas (2008) conjectured, on numerical evidence, that among k-gons the regular k-gon maximises ξ; Theorem 1 settles the case $k = 4$. For triangles the conjecture was proved by Siudeja (2010, acute triangles) and Arbon, Mannan, Psenka, Ragavan (J. Spectral Theory, 2020, all triangles); Arbon (2022) gave a second, computer-assisted proof of the triangle case and proved that the square is a strict local maximiser among quadrilaterals in an explicit but tiny neighbourhood. Endo and Osting (September 2026) proved the Neumann analogue (the square maximises the first nonzero Neumann eigenvalue among convex quadrilaterals of fixed area) by a certified box covering; our

global step follows their architecture, but the Dirichlet ratio needs a sharp *lower* bound on λ_1, which is the main technical difference.

On the existing local result. Arbon's Theorem 1.3 is stated for all quadrilaterals sufficiently close to the square, with the explicit region $(t, s) \in [0, 0.0015]^2$ in his vertex parametrisation. Two steps of the argument do not hold as written. First, equation (3.16) there gives the first-order variation of ξ for a simultaneous perturbation of two vertices as the sum of the first-order variations for the two vertices separately; for the degenerate second eigenvalue this is not valid, because the first-order shift under a sum of perturbations is the smallest eigenvalue of the *sum* of the two 2×2 first-order matrices, not the sum of the smallest eigenvalues. Along the family of isosceles trapezoids the exact first-order matrix vanishes (Lemma 3.2 below), and a finite-element computation is consistent with $\xi \approx 5/2 - 0.70\, t^2$ for the trapezoid with parallel sides $1 \pm t$; the exact first-order term vanishes, so there is no first-order decrease at all, whereas (3.16) predicts a first-order decrease of strength between 0.51 and 1.06 per unit displacement. Second, the bound "$O(t^2) \le 295t^2$" on the second-order remainder in the triangle case (and the analogous constant 336.972 in the quadrilateral case) is inferred from a crude global bound on ξ over a large interval of t; a bound on the size of a remainder term at one value of t does not control its behaviour near $t = 0$. The local statement itself is true (it follows from Theorem 2 below); only the argument needs replacing.

2. Parametrisation and reductions

2.1 Shape parameters

We use the parametrisation of Endo and Osting. For $p = (a, b, c, d) \in \mathbb{R}^4$ let

$$v_1 = (0,0) + \tfrac{1}{2}(a + d - b,\ -a + d + c), \quad v_2 = (1,0) + \tfrac{1}{2}(-a + d + b,\ -a - d - c),$$

$$v_3 = (1,1) + \tfrac{1}{2}(-a - d - b,\ a - d + c), \quad v_4 = (0,1) + \tfrac{1}{2}(a - d + b,\ a + d - c),$$

and Q_p the quadrilateral with these vertices in this cyclic order. Q_0 is the unit square; a is a rectangle mode, b, c are isosceles-trapezoid modes, d a rhombus mode; $|Q_p| = 1 - a^2 - d^2$; Q_p is a nondegenerate convex quadrilateral iff the four cross products $c_i(p) = (v_{i+1} - v_i) \times (v_{i+2} - v_i)$ are positive. Every convex quadrilateral is similar to some Q_p (Endo–Osting, Theorem 2.1), and ξ is invariant under similarities, so it suffices to prove $\xi(Q_p) < 5/2$ for all $p \neq 0$ with all $c_i(p) > 0$. The dihedral group D_4 acts on parameter space by the signed permutations $\sigma_h : (a, b, c, d) \mapsto (a, b, -c, -d)$, $\sigma_v : (a, b, c, d) \mapsto (a, -b, c, -d)$, $\rho : (a, b, c, d) \mapsto (-a, -c, -b, d)$, and $\xi(Q_p)$ is invariant; a fundamental domain is $\{a \ge 0, b \ge 0, c \ge 0\}$.

2.2 Lemma A (near-triangle quadrilaterals)

For a convex quadrilateral with vertices $v_1, \dots, v_4$ write $c_i = (v_{i+1} - v_i) \times (v_{i+2} - v_i) > 0$ (indices mod 4), twice the area of the triangle $v_i v_{i+1} v_{i+2}$.

Lemma A. If $c_i \le \eta\, c_{i+2}$ for some i, then $\xi(Q) \le \frac{7}{3}(1 + \eta)^2$. In particular $\xi(Q) < 5/2$ when $\eta \le 0.035$ (since $\frac{7}{3} \cdot 1.035^2 = 2.4995\ldots$).

Proof. Put $T = \mathrm{conv}(v_{i+2}, v_{i+3}, v_i)$ and let ℓ be the line through v_i and v_{i+2}. The distances of v_{i+1} and v_{i+3} to ℓ are $\delta = c_i/|v_{i+2} - v_i|$ and $h = c_{i+2}/|v_{i+2} - v_i|$, so $\delta/h = c_i/c_{i+2} \le \eta$. Let T' be the image of T under the homothety with centre v_{i+3} and ratio $1 + \delta/h$; it is bounded by the lines $v_{i+3}v_i$, $v_{i+3}v_{i+2}$ and the line parallel to ℓ at distance δ on the far side. Convexity of Q at v_i and v_{i+2} means $(v_i - v_{i+3}) \times (v_{i+1} - v_{i+3}) \ge 0$ and $(v_{i+1} - v_{i+3}) \times (v_{i+2} - v_{i+3}) \ge 0$, i.e. v_{i+1} lies in the closed angle at v_{i+3} spanned by v_i and v_{i+2}; it lies on the far side of ℓ at distance δ, hence on the base of T' (the segment of the line parallel to ℓ at distance δ, cut by that angle). So $v_{i+1} \in T'$ and $Q = \mathrm{conv}(T \cup \{v_{i+1}\}) \subset T'$. Domain monotonicity gives $\lambda_2(Q) \le \lambda_2(T)$ and $\lambda_1(Q) \ge \lambda_1(T') = \lambda_1(T)/(1 + \delta/h)^2$. Hence $\xi(Q) \le (1 + \delta/h)^2\, \xi(T) \le \frac{7}{3}(1 + \eta)^2$ by the triangle theorem (Siudeja 2010; Arbon, Mannan, Psenka, Ragavan 2020). □

2.3 Lemma B (thin convex domains)

Lemma B. Let Ω be a bounded convex planar domain with minimal width w and diameter D, $s = w/D$. Then for every $\theta \in (0,1)$

$$\xi(\Omega) \le \frac{4s^2}{\theta^2} + \frac{1}{(1-\theta)^2}.$$

With $\theta = 1/4$: $\xi(\Omega) \le 64s^2 + 16/9$, which is $\le 2.4178 < 5/2$ for $s \le 1/10$ (optimising θ extends this to $s \le 0.1065$).

Proof. Work with the compact convex set $K = \overline{\Omega}$ (same width and diameter). (i) Ω lies in an open strip of width w, so $\lambda_1(\Omega) \ge \pi^2/w^2$. (ii) Let AB be a diameter of K; put $A = (0,0)$, $B = (D,0)$. Then $\Omega \subset \{0 < x < D\}$ (a point with $x < 0$ would be farther than D from B), and the vertical chords over $x = 0$ and $x = D$ reduce to A and B. Write $\Omega = \{(x,y) : \beta(x) < y < \tau(x)\}$ with τ concave, β convex, $\tau(0) = \tau(D) = \beta(0) = \beta(D) = 0$, and let $H = \tau(x^*) - \beta(x^*)$ be the longest vertical chord. (iii) $H \ge w$: the longest chord of a convex body in direction u equals the radial function $\rho_{\Omega-\Omega}(u)$ of the difference body, the width in direction v equals its support function $h_{\Omega-\Omega}(v)$, and for a centrally symmetric convex body $\min_u \rho = \min_v h$; hence every directional maximal chord is at least the minimal width. (iv) By concavity of τ and convexity of β (both vanishing at 0 and D), on the window $W = [(1-\theta)x^*,\ x^* + \theta(D - x^*)]$ of length θD we have $\tau \ge (1-\theta)\tau(x^*)$ and $\beta \le (1-\theta)\beta(x^*)$. So the closed rectangle $R = W \times [(1-\theta)\beta(x^*), (1-\theta)\tau(x^*)]$, of sides θD and $(1-\theta)H \ge (1-\theta)w$, lies in K; its interior lies in $\operatorname{int} K = \Omega$, and $\lambda_2(\Omega) \le \lambda_2(R) \le \pi^2\big(4/(\theta D)^2 + 1/((1-\theta)w)^2\big)$. Dividing (iv) by (i) gives the claim. □

(The numerical study `explore/thin_REPORT.md` found the best inscribed rectangle fails at $s \approx 0.107$ for every shape family, so Lemma B is essentially sharp for this method; on all 5066 FEM samples the bound exceeds the true ratio, by at least 0.21 for $s \le 0.1$.)

2.4 What remains

By §2.1–2.3 and the local theorem of §3 it suffices to prove $\xi(Q_p) < 5/2$ for p in the fundamental domain $\{a,b,c \ge 0\}$ with all $c_i(p) > 0$, $c_i(p) > 0.035\, c_{i+2}(p)$ for all i, width/diam > 0.1, and $|p|_\infty > 0.04$. Since a convex domain contains the quadrilateral spanned by a diameter and the longest chord perpendicular to it, $|Q| \ge wD/2$, so $D^2 \le 2|Q|/s < 20|Q|$ on the remaining set. Each diagonal is at most D, and the Endo–Osting identity $|v_1 - v_3|^2 + |v_2 - v_4|^2 = 2(1-d)^2 + 2a^2 + 2(1+d)^2 + 2a^2 = 4a^2 + 4(1+d^2)$ (direct computation from the vertex formulas) gives $4a^2 + 4 + 4d^2 \le 2D^2 < 40(1 - a^2 - d^2)$, i.e. $a^2 + d^2 < 9/11$, so $|a|, |d| < 0.9045$. Independently of thinness, convexity bounds b and c: with $S = 1 - a^2 - d^2 = |Q_p| > 0$, $X = b + c$, $Y = b - c$, the four conditions $c_i(p) > 0$ read $|X(1-d) + aY| < S$ and $|Y(1+d) + aX| < S$; inverting $\left(\begin{smallmatrix} 1-d & a \\ a & 1+d \end{smallmatrix}\right)$ (determinant S) gives $|X| < 1 + d + |a|$ and $|Y| < 1 - d + |a|$, hence $|b|, |c| < 1 + |a| < 1.905$. The remaining set therefore lies in the box $a \in [0, 0.905]$, $b, c \in [0, 4.36]$, $|d| \le 0.905$ used as the root region of the covers in §4 (the b, c range is generous; all endpoints are rounded outward in floating point). (Monte Carlo: the remaining set has volume ≈ 0.29 and extent $a \le 0.83$, $b \le 0.85$, $c \le 1.27$, $|d| \le 0.8$.)

3. The local step

3.1 Transported modes, the pencil, and the Kato–Temple bound

Let $\Phi_p : [-1,1]^2 \to Q_p$ be the bilinear map sending the corners to $v_1, \dots, v_4$, with Jacobian J (affine in the reference coordinates (u,v), quadratic in p, $J = \frac{1}{4}$ at $p = 0$). For the square eigenfunctions $g_{mn}(u,v) = 2\sin\frac{m\pi(u+1)}{2}\sin\frac{n\pi(v+1)}{2}$ the transported functions $w = g \circ \Phi_p^{-1}$ lie in $H^2 \cap H_0^1(Q_p)$ and

$$K_{kl} = \int_{Q_p} \nabla w_k \cdot \nabla w_l = \int_{[-1,1]^2} \nabla g_k^{\top} \frac{B}{J} \nabla g_l, \qquad M_{kl} = \int_{[-1,1]^2} g_k g_l\, J, \qquad LL_{kl} = \int_{[-1,1]^2} (Lg_k)(Lg_l)\, J,$$

where $B = \operatorname{adj}(D\Phi_p)\operatorname{adj}(D\Phi_p)^{\top}$ and $Lg = \frac{1}{J}\partial_\alpha\big(\frac{B^{\alpha\beta}}{J}\partial_\beta g\big)$ is the transported Laplacian. With the mode set $\{(m,n) : 1 \le m, n \le N\}$, $N = 8$, we form three test functions

$$\tilde{w}_j = \Big(g_j + \sum_k \sum_{i=1}^{4} C_{jki}\, p_i\, g_k\Big) \circ \Phi_p^{-1}, \qquad j \in \{(1,1), (2,1), (1,2)\},$$

where the fixed rational numbers C_{jki} approximate the first-order perturbation coefficients (they may be any fixed numbers; the choice only affects the size of the certified neighbourhood). Let $\tilde{K}, \tilde{M}$ be the 3×3 Rayleigh–Ritz matrices and $\rho = \tilde{K}_{11}/\tilde{M}_{11}$, $\varepsilon^2 = \widetilde{LL}_{11}/\tilde{M}_{11} - \rho^2 = \|(\Delta + \rho)\tilde{w}_1\|^2/\|\tilde{w}_1\|^2$. For $\beta \le \lambda_2(Q_p)$ with $\beta > \rho$ (Temple's inequality) $\lambda_1(Q_p) \ge L := \rho - \varepsilon^2/(\beta - \rho)$; and $\lambda_2(Q_p) \le \theta_2$, the second eigenvalue of the pencil $(\tilde{K}, \tilde{M})$.

Lemma 3.1 (inertia criterion). Let $A = \tilde{K} - \frac{5}{2}L\tilde{M}$ and let $S = A_{cc} - A_{c1}A_{1c}/A_{11}$ be the Schur complement onto the cluster block $c = \{(2,1),(1,2)\}$. If $A_{11} < 0$ and $\lambda_{\min}(S) < 0$ (i.e. $\det S < 0$ or $\operatorname{tr} S < 0$), then $\theta_2 < \frac{5}{2}L$, hence $\xi(Q_p) < 5/2$. *Proof.* Sylvester's law: the number of pencil eigenvalues $\theta_k \le \sigma$ equals the number of non-positive eigenvalues of $\tilde{K} - \sigma\tilde{M}$. By Haynsworth's inertia additivity, $\operatorname{In}(A) = \operatorname{In}(A_{11}) + \operatorname{In}(S)$, so A has at least two negative eigenvalues; thus $\theta_2 < \frac{5}{2}L \le \frac{5}{2}\lambda_1$ and $\lambda_2 \le \theta_2$. □

Lemma 3.2 (exact structure at the square; sympy, exact arithmetic). With $\lambda_1 = 2\pi^2$, $\lambda_2 = 5\pi^2$: (L0) $K(0) = \operatorname{diag}(\lambda_k)$, $M(0) = I$, $LL(0) = \operatorname{diag}(\lambda_k^2)$; (L1) $\partial_i\lambda_1(0) = 0$ and $\partial_i \operatorname{tr}\big[K_{cc} - 5\pi^2 M_{cc}\big](0) = 0$ for $i \in \{a,b,c,d\}$; the first-order cluster matrices are $\operatorname{diag}(6\pi^2, -6\pi^2)$ for a, $\left(\begin{smallmatrix} 0 & 256/9 \\ 256/9 & 0 \end{smallmatrix}\right)$ for d, and 0 for b and c; (L2) $\partial_i\big[LL_{11} - \lambda_1^2 M_{11} - 2\lambda_1(K_{11} - \lambda_1 M_{11})\big](0) = 0$. Consequently, along every ray $p = te$, A_{cc}, A_{1c} and ε^2 vanish at $t = 0$ exactly, $\varepsilon^2 = O(t^2)$, $S = O(t)$, $\det S = O(t^2)$ and $\operatorname{tr} S = O(t^2)$; the exact first-order term of $\operatorname{tr} S$ vanishes.

3.2 Taylor models and the certificate

All entries K_{kl}, M_{kl}, LL_{kl} are integrals over $[-1,1]^2$ of products of a rational function of (u,v,p) (from B, J and J^{-1}, J^{-2}, J^{-3}) with trigonometric polynomials. Writing $4J = 1 + j_1$ with $|j_1| \le \hat{g}\,|p|_2$ on $[-1,1]^2 \times \{|p|_2 \le \rho_0\}$ (an explicit rational $\hat{g} = s_1 + s_2\rho_0$ from the exact coefficients of j_1), we expand J^{-k} as a truncated binomial series with a geometric tail, keep exact polynomial parts (rational coefficients, degree $\le m$ in p, exact in (u,v)) and bound all dropped terms by $r\,|p|_2^{m+1}$ using $|u^a v^b| \le 1$ and $|p^\gamma| \le \sigma(\gamma)|p|_2^{|\gamma|}$ with $\sigma(\gamma) = \prod_i(\gamma_i/|\gamma|)^{\gamma_i/2}$, which is submultiplicative. Integrals against the trigonometric factors reduce to exact one-dimensional tables evaluated in ball arithmetic (Arb). The Taylor coefficients were validated against direct high-order quadrature (agreement 10^{-8} at $|p| = 0.06$).

Schur complement as a Taylor model. The 3×3 pencil, ρ, ε^2, the Kato–Temple bound L, $A = \tilde{K} - \frac{5}{2}L\tilde{M}$, its Schur complement S, $\det S$ and $\operatorname{tr} S$ are all computed as multivariate Taylor models in p (order m, remainder $r|p|_2^{m+1}$), using the exact vanishing of Lemma 3.2 to delete the degree-0 and degree-1 coefficients that vanish identically (each deleted coefficient is checked to contain 0 in its computed enclosure). Only then are directions substituted. This ordering is essential: substituting interval directions into the pencil first and forming the Schur complement afterwards inflates the enclosures roughly a hundredfold. The shift β is taken from a finite list of levels $36 \le \beta_j \le 49$; for each direction box and s-interval we use the largest level not exceeding the circumscribed-rectangle lower bound for λ_2 (or $36 < 2\pi j_{0,1}^2 \le 2\pi j_{0,1}^2/|Q_p|$, Krahn–Szegő).

The cover. Write $p = sE$ with E on a face of the cube $[-1,1]^4$ and $s = |p|_\infty$. For each box of directions E on a face we certify $s \in (0, 1/25]$; since $|E|_2 \le 2$, $|p|_2 \le 2s \le \rho_0 = 2/25$ (checked in exact rational arithmetic), so the Taylor models apply. On each box and s-subinterval we verify $\det S/s^2 < 0$ or $\operatorname{tr} S/s^2 < 0$ by a second-order centred form in E (exact value and gradient at the box centre, interval Hessian over the box) plus the remainder $r\,s^{m-1}|E|^{m+1}$, together with $A_{11} < 0$, $\tilde{M} \succ 0$ (Gershgorin) and $\beta > \rho$. When a Taylor model is contracted at an order lower than its own, the higher-degree terms are integrated exactly and folded into the remainder.

Theorem 2 (local step). With $N = 10$ (100 transported modes), $m = 12$ (stiffness and mass), $m_{LL} = 14$ (Laplacian), $\rho_0 = 1/10$: for every p with $0 < |p|_\infty \le 1/20$ the criterion of Lemma 3.1 holds, hence $\xi(Q_p) < 5/2$. The certificate consists of 960 direction boxes on the eight faces of the cube (66 s after a 7-minute exact assembly); the configuration $N = 8$, $m = 10$, $\rho_0 = 2/25$ certifies the cube $|p|_\infty \le 1/25$ independently. (Code: `local/engine2.py`, `assemble.py`, `localcert.py`, `mtm.py`, `schur_mtm.py`, `cover2.py`; exact lemmas `lemma_firstorder.py`, `lemma_residual.py`; one-command replays `local/replay_local_v2.sh` (cube 1/20) and `local/replay_local.sh` (cube 1/25).) With $m = 8$ the pipeline certifies $|p|_\infty \le 0.03$ and fails at $\rho_0 = 0.08$; with $m = 12$, $N = 10$ it fails at $\rho_0 = 0.12$.

4. The global step

By §2.4 it remains to treat the compact set $\mathcal{R}$ of parameters p in the fundamental domain $\{a, b, c \geq 0\}$ with Q_p convex, $c_i(p) > 0.035\, c_{i+2}(p)$ for all i, width/diam > 0.1, and $|p|_\infty > 0.049$ (overlapping the cube $|p|_\infty \leq 1/20$ of Theorem 2).

4.1 Test functions and the upper bound

On a box $B = p_0 + \prod_i [-r_i, r_i]$ we use three test functions $u_1, u_2, u_3 \in H^1_0(Q_p)$ depending polynomially on $\delta = p - p_0$, and the Rayleigh–Ritz value θ_2 of the 3×3 pencil $K = (\nabla u_i, \nabla u_j)$, $M = (u_i, u_j)$ bounds $\lambda_2(Q_p)$ from above. *Shell cover:* $u_k = b_p\,(P^0_k + \sum_i \delta_i P^i_k)$ with the edge bubble $b_p = \prod_{k=1}^4 \ell_k$, $\ell_k(x) = (v_{k+1} - v_k) \times (x - v_k)$, fixed polynomials P^0_k (Rayleigh–Ritz eigenvectors at p_0 in a scaled Legendre basis of total degree $n = 8$) and first-order corrections P^i_k (finite differences of the aligned eigenvectors). *Transport cover:* $u_k(p) = \hat{u}_k(\cdot; \delta) \circ \Psi_p^{-1}$, where Ψ_p is affine on the two triangles of a diagonal split of Q_{p_0} with linear parts $A_K(p) = M_K(p) M_K(p_0)^{-1}$ (certified $\det A_K > 0$ on the box, so Ψ_p is a homeomorphism onto Q_p), and $\hat{u}_k$ are bubble polynomials on Q_{p_0}, optionally with first-order corrections. The diagonal is chosen so that a vertex with angle near π is an endpoint of it. Any such choice is admissible; it only affects tightness.

4.2 The lower bound: Lehmann–Goerisch with an exact flux

The Kato–Temple bound used in §3 fails for convex quadrilaterals with an angle near π (for a 157° angle the polynomial residual is too large and the bound loses 40%). We use instead the Lehmann–Goerisch bound with $n = 1$.

Lemma 4.1 (Lehmann–Goerisch, $n = 1$). Let $u \in H^1_0(\Omega) \setminus \{0\}$, $w \in L^2(\Omega)^2$ with $(\nabla f, w) = (f, u)$ for all $f \in H^1_0(\Omega)$ (equivalently $w \in H(\mathrm{div})$, $\mathrm{div}\, w = -u$), and $0 < \beta \leq \lambda_2(\Omega)$ with $A_0 < \beta A_1$. Then $\beta A_2 - A_1 > 0$ and $\lambda_1(\Omega) \geq (\beta A_1 - A_0)/(\beta A_2 - A_1)$.

Proof. Let (φ_k, λ_k) be the orthonormal Dirichlet eigenpairs and $c_k = (u, \varphi_k)$, so $A_1 = \sum c_k^2$ and $A_0 = \sum \lambda_k c_k^2$. The field $w_* = \sum_k c_k \nabla\varphi_k/\lambda_k$ satisfies the same constraint, and $w - w_*$ is L^2-orthogonal to ∇H^1_0, which contains w_*; hence $A_2 = \|w\|^2 \geq \|w_*\|^2 = \sum c_k^2/\lambda_k$. Therefore

$$\lambda_1(\beta A_2 - A_1) - (\beta A_1 - A_0) \;\geq\; \sum_k c_k^2 \Big(\frac{\beta\lambda_1}{\lambda_k} - \lambda_1 - \beta + \lambda_k \Big) = \sum_k c_k^2\, \frac{(\lambda_k - \lambda_1)(\lambda_k - \beta)}{\lambda_k} \;\geq\; 0,$$

since the $k = 1$ term vanishes and $\lambda_k \geq \lambda_2 \geq \beta$ for $k \geq 2$. Finally $A_1 = (\nabla u, w) \leq \sqrt{A_0 A_2}$ gives $\beta A_2 - A_1 \geq A_1(\beta A_1 - A_0)/A_0 > 0$. □

The flux. We take $u = u_1$ and $w = \nabla u_1/\rho_0 + \tau + \mathrm{curl}\, \psi$ with a fixed number ρ_0 (the Ritz value at p_0), $\tau = \varphi\, e$ for the direction e of the split diagonal, where $e \cdot \nabla\varphi = -(\Delta u_1 + \rho_0 u_1)/\rho_0$ and φ vanishes on the other diagonal (a global polynomial, computed exactly by Lagrange interpolation along Duffy lines parallel to e), and ψ a polynomial (least-squares optimal at p_0, with first-order corrections). Then $\mathrm{div}\, w + u_1 = 0$ exactly for every p, and $A_2 = (2\rho_0 A_1 - A_0)/\rho_0^2 + \|\tau + \mathrm{curl}\, \psi\|^2$, which keeps the large part $\nabla u_1/\rho_0$ out of the remainder chain. In the transport cover the reference flux is $\hat{w} = \det A_K\, \hat{\tau} + \mathrm{curl}\, \hat{\psi}$ on each triangle and $w = A_K \hat{w}/\det A_K$ (contravariant Piola map): $\mathrm{div}_x\, w = \mathrm{div}_{\hat{x}}\, \hat{w}/\det A_K = -u_1(p)$, and since $\hat{\tau} \cdot \hat{n} = 0$ on the diagonal the normal fluxes from the two triangles agree, so $w \in H(\mathrm{div}; Q_p)$. The bound only needs the residual in a dual norm, which is why it survives near-straight angles (within 0.1% of λ_1 at 157°).

The shift $\beta \leq \lambda_2$. Either a constant valid on the whole box, the maximum of: Krahn–Szegő $2\pi j^2_{0,1}/\max_B |Q_p|$; enclosing rectangles in fixed directions, $\pi^2 \min(4/A^2 + 1/B^2,\, 1/A^2 + 4/B^2)$; and the Andrews–Clutterbuck gap $\lambda_2 \geq \lambda_1 + 3\pi^2/D^2$ with a lower bound for λ_1 from rectangles or Hersch's inequality — or (transport cover) a Taylor model $\beta(p) \leq \lambda_2(Q_p)$ valid pointwise (Krahn–Szegő with the exact area $1 - a^2 - d^2$, rectangles with affine side bounds, the gap bound, and a bootstrap that feeds the Lehmann–Goerisch value back into the gap bound). For elongated shapes, where no available β exceeds λ_1, a *domain-monotonicity mode* replaces L by a scalar $L_0 \leq \lambda_1$ (enclosing rectangles); only 12 boxes of the two covers use it.

4.3 The criterion and its Taylor models

With $\tilde{A} := K(\beta A_2 - A_1) - \kappa(\beta A_1 - A_0)M = (\beta A_2 - A_1)(K - \kappa LM)$ (or $\tilde{A} = K - \kappa L_0 M$ in the domain-monotonicity mode), Lemma 3.1 applies verbatim: if $\tilde{A}_{11} < 0$ and $S' = -\tilde{A}_{11}\tilde{A}_{cc} + \tilde{A}_{c1}\tilde{A}_{1c}$ has a negative eigenvalue (detected by $S'_{11} < 0$, $S'_{22} < 0$, $\operatorname{tr} S' < 0$ or $\det S' < 0$), then $\tilde{A}$ has two negative eigenvalues (Haynsworth), and $\theta_2 < \kappa L \le \kappa\lambda_1$. The side conditions $\beta A_1 - A_0 > 0$ and $M \succ 0$ are certified as well ($\beta A_2 - A_1 > 0$ then follows). The theorem uses $\kappa = 5/2$; every certificate is checked at $\kappa = 5/2$ first, with strict inequalities.

All quantities are exact integrals of polynomials: the triangles of the split are integrated with a Duffy–Gauss–Legendre product rule with $m = n + 6$ points, exact up to total degree $2n + 10$, which covers every integrand; nodes and weights are Arb balls. In the shell cover the base quantities are order-3 Taylor models in $t = \delta/r \in [-1, 1]^4$ (vertices and nodes move linearly; fixed polynomials are expanded at the moving nodes with a rigorous tail; every product folds the dropped coefficients into the remainder). In the transport cover all p-dependence sits in $A_K(p)$ and δ: K, M, A_2 are explicit rational functions of p built from reference integrals on Q_{p_0} that are quadratic in δ and computed once per box in ball arithmetic; denominators are cleared with $\det A_1 \det A_2 > 0$, and the resulting polynomials are order-4 Taylor models with exact full products (coefficients of degree ≥ 5 folded into the remainder). Since the criterion is homogeneous in (K, M, A_2), the base models are first multiplied by a certified positive polynomial approximating $1/M_{11}$, which removes the common dilation variation without changing any sign. Ranges over the box are bounded in ball arithmetic.

4.4 The cover

Near-square shell (fixed test functions). With fixed polynomial test functions (degree $n = 8$, first-order corrections in p), order-3 Taylor models and the criterion above, the shell $0.049 < |p|_\infty \le 0.225$ of the fundamental domain is covered by 8216 certified boxes (8209 Lehmann–Goerisch, 7 with a scalar domain-monotonicity bound $L_0 \le \lambda_1$), with no failures, in 1171 s on 8 threads; an independent replay recomputes every certificate from the log and checks coverage (certified volume equals the region's volume). The smallest certified margin is $5/2 - \xi_{\text{bound}} = 2.36 \times 10^{-5}$.

Theorem 3 (near-square region). For every convex quadrilateral Q_p with $0 < |p|_\infty \le 0.225$, $\xi(Q_p) < 5/2$.

Remaining region (transported test functions). Fixed test polynomials lose accuracy across a box when a vertex angle is close to π (the eigenfunction's corner behaviour moves with the vertex); the pilot predicted $\ge 10^7$ boxes. We therefore transport the box-centre test functions with the vertices, $u_k(p) = \hat{u}_k \circ \Psi_p^{-1}$ with Ψ_p affine on the two triangles of the v_1v_3 split, and the flux by the contravariant Piola map, $w = A_K\hat{w}/\det A_K$ with $\hat{w} = \det A_K\, \hat{\tau} + \operatorname{curl}\hat{\psi}$, where $\hat{\tau}$ is the antiderivative of $\hat{u}_1$ along the diagonal (so $\hat{\tau} \cdot n = 0$ on the diagonal). Then $\operatorname{div} w = -u_1(p)$ exactly for every p, and all p-dependence sits in the two 2×2 matrices $A_K(p)$, so every quantity is an explicit low-degree rational function of p built from reference integrals computed once per box.

The cover starts from the a-priori box of §2.4 cut at $a, b, c = 1/5$ and $d = \pm 1/5$ (24 roots; the root $[0, \frac{1}{5}]^3 \times [-\frac{1}{5}, \frac{1}{5}]$ lies in the region of Theorem 3). A box is either certified, bisected, or discarded because one of the following holds at every point of the box: E2 (Lemma A, $c_i \le 0.035\, c_{i+2}$ for a fixed i), E4 ($c_i \le 0$ for a fixed i: not convex), SHELL (inside the region of Theorem 3), E3T (width/diam $\le 2/5$, decided by an exact sufficient test on one edge and one vertex pair), or a recursive combination of these on sub-boxes (COMPOUND). The run produced 100 342 records forming 24 complete bisection trees with 50 183 leaves:

leaves	count	share of the root volume
certified (Lehmann–Goerisch 31 421; domain monotonicity 5)	31 426	0.45 %
E2 (Lemma A)	13 933	2.50 %
E4 (not convex)	138	4.04 %
E3T (width/diam $\le 2/5$; Lemma B′ of §4.5)	1 568	0.36 %
COMPOUND (1 010 of them use E3T on some sub-box)	3 117	92.62 %
SHELL	1	0.01 %

There were no failures and no errors. Every certificate holds at $\kappa = 5/2$ with strict inequalities; on a ladder of trial values the smallest certified κ has median 2.31 and minimum 1.66, and about 1 % of the boxes certify only at $\kappa = 5/2$ itself. The run took about 5 200 s on 8 threads of a shared machine (it was interrupted once by a system restart and resumed from its log; the driver skips finished leaves and re-expands logged splits, so no box was certified twice or lost).

Independent checks. (i) A separate exact-arithmetic program (`verify/xcover.py`) rebuilds the 24 roots and the bisection trees from the log, checks that every leaf's box equals the reconstructed one (so the leaves tile the root box exactly: exact rational volume fractions sum to 1), and re-decides every discard with its own exact polynomial bounds and recursive refinement: 0 problems. The same program confirms the shell cover of Theorem 3 (8 216 certified leaves and the central cube; volume sum exactly 1). (ii) A clean replay of the frozen log, bound to its hash and to the hashes of the verifier sources, re-verified all 31 426 certificates and re-checked all 18 756 exclusion and compound leaves with 0 failures (§6). (iii) A second implementation of the transported certificate re-certifies a stratified random sample of certified boxes (§6).

4.5 Lemma B′: the thin band $1/10 < \mathrm{width}/\mathrm{diam} \le 2/5$

Lemma B′. Every convex quadrilateral Q with $1/10 < s := \mathrm{width}(Q)/\mathrm{diam}(Q) \le 2/5$ satisfies $\xi(Q) < 5/2$.

Thin shapes defeat the two-dimensional certificates of §4.4 for a structural reason: $\lambda_1 \approx \pi^2/w^2$ varies by $\pm 8r/w$ over a parameter box of radius r, so any box-constant eigenvalue bound loses that factor twice. We therefore normalise by the diameter and use one-dimensional bounds along it, which follow the $1/w^2$ scaling.

Charts. Normalise a diameter pair of vertices to $(0,0),(1,0)$ (all six vertex distances are then ≤ 1). If the pair is a diagonal (chart I), the other vertices are $P_1 = (x_1, -ty_3)$ and $P_3 = (x_3, y_3)$ with $0 < t \le 1$, and by the reflections $y \mapsto -y$, $x \mapsto 1-x$ we may assume $x_3 \le 1/2$; the triangle $P_0P_2P_3$ has longest side 1, so $y_3 \le s \le (1+t)y_3$, and the band forces $(y_3, t, x_1, x_3) \in [1/20, 2/5] \times [0,1] \times [0,1] \times [0,1/2]$. If the pair is a side (chart II), the other vertices are $P_2 = (x_2, y_2)$, $P_3 = (x_3, ty_2)$ with $0 < t \le 1$ after a reflection, $x_3 < x_2$, and $s = y_2$ exactly, so $(y_2, t, x_2, x_3) \in [1/10, 2/5] \times [0,1]^3$. All cross products c_i are multilinear in these coordinates, so their ranges over a box are attained at its corners.

Lower bound for λ_1 (one-dimensional reduction). If $h_Q(x)$ is the length of the slice $Q \cap \{x = \mathrm{const}\}$ and $0 \le V \le \pi^2/h_Q^2$, then $\lambda_1(Q) \ge \mu_1\big(-\frac{d^2}{dx^2} + V\big)$ with Dirichlet conditions on $[0,1]$: apply the one-dimensional Poincaré inequality on each slice to $\varphi \in C_c^\infty(Q)$ and note that $v(x) = \|\varphi(x,\cdot)\|_{L^2}$ satisfies $|v'| \le \|\partial_x \varphi(x,\cdot)\|$. For a parameter box we take a polyline $h_{\mathrm{out}} \ge h_Q$ valid for every Q in the box and the cellwise-constant minorant $V = \pi^2/\max_{\mathrm{cell}} h_{\mathrm{out}}^2$; $\mu_1 > E$ is certified by Sturm oscillation (the solution of $-v'' + (V-E)v = 0$, $v(0) = 0$, $v'(0) = 1$ stays positive on $(0,1]$), using exact transfer matrices in ball arithmetic.

Upper bound for λ_2. For an inner domain $Q_{\mathrm{in}} = \{a(x) < y < b(x)\}$ contained in every Q of the box (a, b polylines) and $u = f(x)\sin(\pi(y-a)/h)$, $h = b - a$, $f(0) = f(1) = 0$:

$$\int |\nabla u|^2 = \int_0^1 \Big[\tfrac{h}{2} f'^2 + \tfrac{h'}{2} f f' + \tfrac{\pi^2}{h} f^2 \Big(\tfrac{1}{2} + \tfrac{a'b'}{2} + h'^2\big(\tfrac{1}{6} + \tfrac{1}{4\pi^2}\big)\Big)\Big] dx, \qquad \int u^2 = \int_0^1 \tfrac{h}{2} f^2 \, dx.$$

With two piecewise-linear functions f_1, f_2 the 2×2 Gram matrices N, D are enclosed exactly element by element (closed forms with logarithms, or rigorous series when the element is nearly uniform).

Certificate. If $\mu_1 > E$ is certified and $\frac{5}{2}E\,D - N \succ 0$ (leading minors positive as balls), then $\lambda_2(Q) \le \lambda_2(Q_{\mathrm{in}}) < \frac{5}{2}E < \frac{5}{2}\lambda_1(Q)$ for every valid Q in the box.

Exclusions (each decided exactly on the whole closed box): Lemma A; $s \le 1/10$; $s > 2/5$ (using the exact formula $w = \min_i \max(c_i, c_{i+3})/|v_{i+1} - v_i|$ for the width of a quadrilateral); and boxes containing no valid chart point.

Review and repair. An adversarial review found that the first implementation formed products of nodal values in floating point before converting them to balls, so the Gram enclosures did not contain the exact forms (errors near 10^{-15} against radii near 10^{-34}), and that one series remainder radius was computed without directed rounding. Both were repaired (all inputs promoted to balls before any arithmetic; the remainder computed with directed upper bounds), new tests check strict containment of every element integral against exact rational references (the old assembly fails them on all 72 test meshes), and every leaf was re-certified with the repaired code: no leaf changed status or needed further subdivision.

The cover. Chart I: 45 169 leaves (36 696 certified, 8 094 with $s > 2/5$, 141 with $s \le 1/10$, 110 Lemma A, 128 invalid); chart II: 8 048 leaves (7 572 certified, 161 Lemma A, 315 invalid); no box needed refinement below width 2^{-24}; largest certified value of $R_2/(\frac{5}{2}E)$: 0.9999985 and 0.99997 (a box is certified as soon as the strict inequality holds). The replay of the re-certified logs recomputes every box from its path, re-derives every exclusion and re-runs every certificate (583 s on three threads, 0 errors); it fails closed on unknown, duplicate, malformed or pending records and binds each result to hashes of its log record and of the verifier sources; twenty deliberately corrupted logs were all rejected. *Independent checks:* a separately written exact program (`verify/xbandcover.py`) rebuilds both tilings from the paths (exact volume), and re-derives all 8 949 exclusions with its own predicates (0 problems); a separately written implementation of both one-dimensional bounds (`verify/xband.py`; validated against two-dimensional finite elements on 40 random band shapes, where it brackets both eigenvalues with margins of 1–10 % and 0.07–6 %) re-certifies a random sample of 200 certified boxes per chart of the re-certified logs: all 400 certificates are reproduced (382 with its default mesh, the other 18, all with logged margins above 0.997, after refining its own one-dimensional meshes). An adversarial review also found a defect in this independent checker (its outer slice profile was too small for boxes touching $x = 0$); it was repaired and the whole sample re-run before the numbers above were obtained. Every receipt records SHA-256 hashes of the checked logs and of the checker sources.

4.6 Convex quadrilaterals

Theorem 1′. For every convex quadrilateral Q, $\xi(Q) \le 5/2$, with equality if and only if Q is a square.

Proof. Let Q be a convex quadrilateral that is not a square. By Endo–Osting (Theorem 2.1) Q is similar to some Q_p with all $c_i(p) > 0$, and applying an element of D_4 we may take p in $\{a, b, c \ge 0\}$; $p \ne 0$. If $c_i(p) \le 0.035\, c_{i+2}(p)$ for some i, Lemma A gives $\xi(Q) < 5/2$. If width/diam $\le 1/10$, Lemma B gives $\xi(Q) \le 2.418$; if $1/10 <$ width/diam $\le 2/5$, Lemma B′ gives $\xi(Q) < 5/2$. If $0 < |p|_\infty \le 1/20$, Theorem 2 applies, and if $|p|_\infty \le 0.225$, Theorem 3. Otherwise p lies in the a-priori box of §2.4 (because width/diam $> 1/10$), hence in a leaf of the transported cover; that leaf is not a discard of type E2, E3T, E4 or SHELL covering p (those cases were treated above or are impossible for convex Q_p), so p lies in a certified box and $\xi(Q_p) < 5/2$. (For COMPOUND leaves the same reasoning applies to the sub-box containing p.) For the square $\xi = 5\pi^2/2\pi^2 = 5/2$. □

5. Non-convex quadrilaterals (darts)

Theorem 5. For every simple non-convex quadrilateral Q, $\xi(Q) < 5/2$.

5.1 Setting

A dart has one reflex vertex D; call the opposite vertex B and the others A, C (boundary $A \to B \to C \to D \to A$). The diagonal BD lies inside Q, $Q = \mathrm{int}(ABD \cup CBD \cup BD)$, its convex hull is the triangle $T = ABC$, and the notch is ACD. By a similarity, $B = (0,0)$, $D = (1,0)$, $A = \rho_A(\cos b_A, \sin b_A)$, $C = \rho_C(\cos b_C, -\sin b_C)$ with $b_A, b_C \in (0, \pi)$ (the *BD chart*); the reflection in BD exchanges A and C, so we may assume $b_A \le b_C$. The line AC meets the ray BD at abscissa $f = \sin\theta_B/(\sin b_A/\rho_C + \sin b_C/\rho_A)$, $\theta_B = b_A + b_C$, and Q is a dart with reflex vertex D iff $\theta_B < \pi$ and $f > 1$; the notch depth $\beta = \mathrm{dist}(D, AC)/\mathrm{dist}(B, AC)$ equals $1 - 1/f$. Numerically (finite elements, adversarial search), the supremum of ξ over darts is 7/3, approached only along three degenerate curves: shallow notches over an equilateral hull ($\beta \to 0$), and an equilateral CBD with a vanishing arm ABD (a thin spike, or $A \to D$). The shape space is non-compact (arms of unbounded length, deep notches $\beta \to 1$), so the proof combines analytic lemmas for the degenerate and unbounded regions with certified covers of compact families.

5.2 Analytic results (complete proofs in `darts/LEMMAS.md`)

- **Shallow notch** (Proposition 5.1): if $\beta \le 0.0339$ then $\xi(Q) \le \frac{7}{3}(1-\beta)^{-2} < 5/2$ (homothety of the hull at B and the triangle theorem).

- **Collapse collar** (Lemma 3.1): if A is within 0.035 times the inradius of CBD from the segment BD (or $A \leftrightarrow C$), then $\xi(Q) \le 2.4995$.
- **Thin wedge at** B (Lemmas 5.1, 5.3): $\tan b_A + \tan b_C \le 1/10$ gives $\xi \le 2.4223$; $\theta_B \le 0.1772$ gives $\xi \le 2.4892$ (sector comparison, Bessel zeros certified in ball arithmetic). **Flat hull** (Lemma 5.2): hull angles at A and C at most $\arctan(1/10)$ give $\xi \le 2.4356$. **Thin stem** (Lemma 12.1): if the vertical chords of Q over BD are at most $0.05\,|BD|$, then $\xi \le 2.4356$.
- **Long arms** (Lemmas 6.1, 9.1–9.5): cutting an arm by a segment and imposing a Neumann condition on the cut (Dirichlet–Neumann bracketing) bounds λ_1 below by the mixed eigenvalue of a bounded core and the chord bound π^2/ℓ^2 of the cut-off tip, uniformly in the arm length; together with domain monotonicity for λ_2 this reduces arms of any length to compact families. A closed-form bound (Lemma 9.5) disposes of very long arms making an angle between 40° and 140° with BD.
- **One-dimensional bounds for dart families** (Lemmas 11.1–11.3): for any open set, $\lambda_1 \ge \inf \sigma\big(-\frac{d^2}{dx^2} + \pi^2/\ell(x)^2\big)$, where $\ell(x)$ is the length of the *longest component* of the slice at x; slab-wise upper bounds of ℓ valid for whole families of darts (including arms of unbounded length, via half-strip containment and exponential tails), Sturm shooting in ball arithmetic (on unbounded intervals with a decaying-exponential start and an Allegretto–Piepenbrink argument), and a P2 Rayleigh–Ritz bound on an *inner dart* contained in every dart of the family give a certificate that is uniform in the arm lengths.

5.3 Certificates for boxes of darts

Three certificates are used, each valid for every dart of a parameter box. (a) *Finite elements* (§8, §10 of `LEMMAS.md`): a Crouzeix–Raviart discretisation on an affine image of a fixed reference mesh with Liu's guaranteed lower bound $\lambda_k \ge \lambda_{k,h}/(1 + 0.1893^2 h^2 \lambda_{k,h})$, a Temple step whose shift $\sigma \le \lambda_{2,h}$ is certified by one verified LDL^T inertia count, transfer between parameter points by a strain bound, and a conforming P2 Rayleigh–Ritz bound for λ_2. (b) *BO2*: the one-dimensional lower bound of §5.2 for λ_1 and a P2 Ritz bound for an inner dart. (c) *INC* (Lemma 11.4): an outer dart containing every dart of the box and an inner dart contained in all of them, with a point finite-element lower bound for the outer and a P2 upper bound for the inner one. All inequalities are decided in ball arithmetic; box geometry is exact.

5.4 The decomposition and the covers

Write $t = \tan b_A$. The dart condition in polar form (Lemma 13.1 of `LEMMAS.md`) gives two facts used throughout: $1 - \beta = (\sin b_C/\rho_A + \sin b_A/\rho_C)/\sin\theta_B$, and every dart has $\rho_A > \sin b_C$ and $\rho_C > \sin b_A$.

Theorem 5.4 (Theorem 13.2 of `LEMMAS.md`). Every dart with $b_A \le b_C$ (so $b_A < \pi/2$) lies in at least one of the following sets, and each set is covered by a result of §5.2 or by a certified run:

1. $\theta_B \le 0.1772$: Lemma 5.3.
2. $\theta_B \ge 174.29°$: Lemma 5.2 (the hull angles at A and C sum to at most 5.71°).
3. $\beta \le 0.0339$: Proposition 5.1.
4. every vertical chord of Q over BD has length at most 0.05: Lemma 12.1.
5. *Collar*, $t < 0.02$, every ρ_A: run SP_col for $\rho_C \in [e^{-4}, e^3]$, together with the earlier finite-element run kf_collar for C in the sector $38° \le b_C \le 100°$, $0.5 \le \rho_C \le 1.8$; run SF_col for $\rho_C \ge 20$; Lemma 12.1 for $\rho_C < e^{-4}$ (the stem chords are then below $0.02 + e^{-4} < 0.05$).
6. *Spike*, $0.02 \le t \le 3/10$ and $a_x \ge 1$, every ρ_A: run SP_spk (with kf_spike in the same sector) for $\rho_C \in [e^{-4}, e^3]$, and run SF_spk for $\rho_C \ge 20$. For $\rho_C < e^{-4}$ there is no dart, since $\rho_C > \sin b_A > 0.0199$.
7. All other darts. If $\rho_A < e^{-2.5}$ then $\sin b_C < \rho_A < 0.083$, so $b_C < 4.8°$ (and $\theta_B \le 2b_C$ is in set 1) or $b_C > 175.2°$ (set 2). If $\rho_C < e^{-4}$ there is no dart. The polar box $\rho_A \in [e^{-2.5}, e^3]$, $\rho_C \in [e^{-4}, e^3]$ is run PP. Darts with an arm of length $\rho \ge 20$ are runs PF, FP and FF; the polar and far grids overlap because $e^3 > 20$.

The certificates of the collar and spike runs hold for every arm length $|BA|$ at once. Lemmas 9.1–9.4 cut the arm BA and put a Neumann condition on the cut, and the BO2 bound treats the spike as a far arm. The far runs bound λ_1 from below with exponential tails (Lemma 11.3(ii)), uniformly in $\rho \ge 20$. Inside PP, a box with

exactly one arm of $\rho \geq 4$ is certified in the oblique-cut chart of Lemma 9.6 (FARL), and two compact regions near the critical configurations are delegated to earlier finite-element runs in Cartesian BD coordinates. These are `shallow_crit`, a box around shallow notches over an equilateral hull, and `main_cbox`, which takes C in a box around the equilateral apex with $t \geq 0.3$ and $a_y \leq 1.5479$. Two further runs, `crit_collar` and `crit_spike_X1`, cover the spike charts for C in the same box. Every other box of a run is either certified or discarded by a re-decided predicate naming one of the sets 1–7, "not a dart" ($\theta_B \geq \pi$ on the box) or "mirror" ($b_C \leq b_A$ on the box).

Exactness of the grids. A box is stored as floats (p_0, r) and denotes the closed box $[p_0 - r, p_0 + r]$ in exact reals; every leaf, skipped cell and chart box is checked to have exactly representable ends before any predicate or certificate uses it. Grid cells whose nominal ends are not dyadic (the spike cells $0.02 \cdot 1.5^i$) are encoded outward. The processed roots of the earlier runs, stored with nearest-rounded encodings, are represented by their outward dyadic snapping, which their leaves tile exactly. A separate accounting program checks in rational arithmetic that the represented boxes of every family cover its parameter ranges. Decimal thresholds enter every predicate as exact decimals compared with directed ball endpoints.

Statistics. The re-certified family logs (all leaves recomputed with the repaired verifier; 4 workers, 2.69 h) and the earlier runs:

run	region (Theorem 5.4)	leaves	discards	BO2	INC	FEM
PP	set 7, polar arms	306 017	84 142	126 002	51 382	44 491 (27 861 FARL)
SP_col	set 5, $\rho_C \in [e^{-4}, e^3]$	43 078	2 375	37 722	—	2 981
SP_spk	set 6, $\rho_C \in [e^{-4}, e^3]$	62 671	2 655	52 228	—	7 788
PF	set 7, $\rho_C \geq 20$	8 545	2 924	5 621	—	—
FP	set 7, $\rho_A \geq 20$	29 339	2 201	17 420	—	9 718
FF	set 7, both arms ≥ 20	540	141	399	—	—
SF_col / SF_spk	sets 5 / 6, $\rho_C \geq 20$	393 / 996	23 / 91	370 / 905	—	—
earlier runs (6 logs)	critical regions, KF sector	11 270	1 042	—	—	10 228
total		**462 849**				

Every leaf of PP, SP_spk, PF, FP, FF and the far families re-certified at its original box. In SP_col, 232 finite-element leaves whose t-range exceeds the collar chart's t_1 were subdivided and re-certified (+1 193 leaves); no leaf failed.

Replay. All dart logs, the earlier runs included, were replayed from scratch by the aggregate driver `replay_all.py` in proof mode. The replay is fail-closed: unknown, duplicate, malformed or pending records are errors, the tiling of every root is checked exactly, every discard is re-decided and every certificate is recomputed. Each result is bound to the SHA-256 of its log record and to a binding hash. That hash covers every decision source, every cached chart, the manifest bytes, the family and the runtime identity (interpreter, numerical libraries, floating-point environment). The driver runs the analytic gates (the certified tables of Lemmas 5.3 and 9.5), the chart and accounting checks and the rigour lint before the replays. It validates every receipt record of every shard against the bound log and writes an append-only, hash-chained receipt. Result: PASS. All 462 849 leaves of the 14 logs were recomputed or re-decided, 2 shards per log, with no failure (5.63 h). The 33-record receipt has final digest `35d4a6fbdbbdafac...` (receipt file SHA-256 `a2759b4fcd03361d...`, pin `7e564f3f...`). It is verified by `verify/xreceipt.py`, an exact state machine checked against the reviewed pin and the runtime anchors of the preceding structure-only run.

Independent coverage check. `verify/xdart.py`, written separately, samples darts and checks that each lies in a lemma region (re-decided with exact or directed arithmetic) or in a box of a certified leaf of the logs (no enlargement). It uses 160 000 samples: uniform in each run's parameters, concentrated near the boundaries of the lemma regions and of the grids (ten kinds of boundary), and 20 000 with $\log \rho$ up to 40. On the pre-repair logs no sample was uncovered, and on the re-certified logs none was either (XDART PASS, 267 s; receipt with the SHA-256 of every log it read). This check is evidence of correct bookkeeping, not part of the proof; the exact tiling and accounting are.

6. Verification, reproducibility and method statement

Rigour conventions. Floating point is used only to choose test functions, fluxes, correction coefficients and box splits; every inequality that enters a proof is evaluated in ball arithmetic (Arb, through python-flint and Arblib) or with outward-rounded intervals, with exact rational box endpoints. Every Taylor model carries an explicit remainder; truncating to a lower order folds the dropped terms into the remainder.

Replays. Every certificate log is replayed from scratch by a fail-closed script: unknown, duplicate, malformed or pending records are errors; every box is recomputed from its log record (path or centre and radius); every exclusion is re-decided; every certificate is recomputed; and every result is bound to SHA-256 hashes of its log record and of the verifier sources. Local step: `local/replay_local_v2.sh`. Shell cover: exhaustive fail-closed replay of all 8 217 leaves (coverage exact; 8 216 certificates re-verified, 0 failures; 2 888 s on three threads). Transport cover: clean replay of all 50 182 non-shell leaves of the frozen log (SHA-256 `cb187ebb...`): all 31 426 certificates re-verified and all 18 756 exclusion and compound leaves re-checked, 0 failures, smallest certified margin $5/2 - \kappa^* = 5.05 \times 10^{-6}$ (7 481 s on three threads; receipt with result hash `c9fa7f8b...`). Lemma B': 44 268 certificates and 8 949 exclusions, 0 errors. Darts: fresh fail-closed replay of all 14 logs, 462 849 leaves, PASS (§5.4).

Independent programs (separately written, in Python with exact rationals and python-flint balls). (i) `verify/xcover.py` rebuilds the bisection trees of the shell and transport covers from their logs, checks that every leaf box equals the reconstructed one (so the leaves tile the root boxes exactly) and re-decides every discard with its own exact predicates and refinement: both covers pass (0 problems). (ii) `verify/xcert.py` is a second implementation of the transported certificate (exact rational reference integrals, its own choice of split diagonal, constant and pointwise Krahn–Szegő shifts, range bounding by Taylor shifts of exact polynomials); on a stratified random sample of 60 certified transport leaves it re-certifies 52, by its own subdivision of the box and its own exact exclusions (Lemma A, non-convexity) for sub-boxes that need none; the remaining 8, all among the largest boxes, exceed the per-box work budget of this simpler implementation (it has no first-order test-function corrections), and none fails. (While building it we found and fixed two defects of our own in it: a fixed choice of split diagonal, degenerate for near-triangles, and a criterion demanded at non-convex points of boxes that straddle the convexity boundary.) (iii) `verify/xbandcover.py` and `verify/xband.py` re-check Lemma B' (§4.5). (iv) `verify/xdart.py` checks that the dart decomposition is covered: 160 000 sampled darts, including samples concentrated at every boundary of the lemma regions and grids and arms up to e^{40}, each lie in a lemma region or a certified box of the final logs (0 uncovered). The dart certificates themselves have no second, independently written implementation. Their evidence is the fail-closed replay of §5.4 and the adversarial reviews. All receipts record the SHA-256 hashes of the inputs and of the checker sources.

Adversarial reviews. A second model (GPT-5.6 "Sol", OpenAI), prompted to refute, reviewed every part. (a) Lemmas A and B and the local step: two defects in the certificate code (an endpoint rounded to 0.039999999999; Taylor terms dropped when lowering the order without bounding them), both corrected before the replay reported here; Lemmas A and B, the Kato–Temple and inertia arguments and the analysis of Arbon's local argument were confirmed. (b) The global covers: the mathematics of both certificates (conformity of the transported test functions, exact divergence and normal continuity of the Piola flux, the shifts β, the inertia criterion, the Taylor-model algebra, coverage and the a-priori box) passed; the review found assurance defects (a replay cache keyed only by path, status handling that did not fail closed — also in our independent coverage checker — and the then-unfinished independent recertification), all repaired before the receipts above. (c) Lemma B': a fatal implementation defect (floating-point products formed before conversion to balls in the Gram matrices), an unproved floating-point remainder radius, and a bug in our independent checker (§4.5); all repaired and the cover re-certified. (d) Darts, in four rounds. Round 1 refuted the first package. It found nearest-float conversions in

rigorous decisions (the finite-element Schur test, strain factors, arctangent and cotangent endpoints of spike and FARL boxes, the FARL tail and the BO2 projection range), 81 uncovered slivers between skipped spike grid cells, an unguarded NaN path in a Taylor-model range, a replay not bound to its inputs, gaps in Lemmas 5.1, 5.2, 9.3 and 9.4, and a missing lemma for oblique cuts of long arms (now Lemma 9.6). All were repaired and every family leaf re-certified. Round 2 cleared the numerics and Lemma 9.6, and rounds 2 and 3 found defects only in the replay harness: shard-count and quick-mode validation, time-of-check snapshots, the floating-point environment, the aggregate receipt and reuse of earlier replays. These were repaired and tested by probes that corrupt each input in turn. Rounds 4 and 5 found no defect in the certificates and none in the kernel. They found gaps in the provenance of the runs: inputs that a concurrent honest edit could change between the start and end checks, an interpreter that could drift between logs, and a receipt validator that accepted a run with no terminal record. These were closed operationally. Every pinned file and the directories holding them were made immutable for the duration of the runs. The reviewed pin digest became a launch precondition, and the runtime identity was anchored to an earlier run. An independently written receipt verifier (`verify/xreceipt.py`, an exact state machine over the 33 receipt records) checks the completed run. It was itself mutation-tested: 27 re-chained forgeries were rejected. Before the final replay, a recomputation of a sample containing every certificate method found a defect introduced by the round-4 changes. A local variable in the replay script shadowed the module used by the end-of-run checks, so every shard containing finite-element leaves crashed after its last leaf. The certificates themselves were unaffected. The variable was renamed, the pin regenerated (it differs only in that file), and the sample re-run on shards containing 234, 174, 106 and 16 finite-element leaves of PP, SP_col, SP_spk and FP, with no failure. The structural identities of Lemma 3.2 are verified symbolically (sympy), and the Taylor coefficients of the local step agree with direct high-order quadrature.

Method statement. The problem selection, proofs, code and computations were produced by AI systems (Anthropic Claude, with an OpenAI model used as an adversarial reviewer) operating autonomously; no human contributed mathematical content. The author line is "Anonymous" accordingly.

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